PROJECT REPORT — ArogyaMitra

PROJECT TITLE

ArogyaMitra — AI-Powered Fitness & Wellness Web Platform

PROJECT DESCRIPTION

System Overview

ArogyaMitra is a full-stack AI-driven fitness and wellness application designed to generate personalized workout plans, weekly nutrition schedules, smart grocery lists, and provide AI-powered coaching through an interactive dashboard.

The system features a modern React + TailwindCSS frontend, a Node.js Express backend, Groq API / LLM-based fitness coaching, and a MongoDB database for storing user profiles and health data.

Problem Domain

Traditional fitness apps provide static plans that lack personalization and adaptability. Users must rely on manual tracking, and accurate nutrition/workout planning requires expert guidance. ArogyaMitra solves this by using AI to create adaptive recommendations tailored to a user's metrics, goals, allergies, medical history, and daily progress.

Core Technology Used

- Frontend: React, TailwindCSS
- Backend: Node.js, Express
- AI Services: Groq API, Python micro-services (when applicable)
- Database: MongoDB
- External Integrations: YouTube API (exercise videos), Spoonacular API (nutrition)

Key Capabilities

- AI-generated weekly workout plans
- Weekly meal plans with macro breakdowns
- Auto-generated categorized grocery list
- AI wellness coach "AROMI"
- Health profile storage
- Interactive dark-themed dashboard

Target Users

- Fitness enthusiasts

- Individuals seeking personalized health guidance
- Beginners needing structured routines
- Users with medical/dietary restrictions
- Busy professionals needing simplified meal/workout planning

Real-world Relevance

The system can replace multiple separate tools: calorie trackers, meal planners, workout apps, and chat-based AI assistants—offering a unified, automated fitness ecosystem.

APPLICATION SCENARIOS / USE CASES

1. End-User Fitness Application

Users track daily workouts, access exercise videos, analyze macros, generate grocery lists, and adjust nutrition based on AI inputs.

2. Personalized AI Wellness Coaching

AROMI assists users in real time—answering queries, modifying plans, explaining exercises, or offering nutrition/medical-safe advice.

3. Corporate Wellness Programs

Organizations can use ArogyaMitra to encourage employee fitness engagement and track wellness statistics.

4. Educational / Training Use

Fitness trainers or students can study AI-based health planning and test protocols for different body types and goals.

TECHNICAL ARCHITECTURE OVERVIEW

High-Level Architecture

- Frontend (React + Tailwind)
- |
- | REST API Requests
- v
- Backend (Node.js, Express)
- |
- | DB Queries / AI Requests
- v
- MongoDB Database External APIs:
- - Groq AI
- - YouTube API
- - Spoonacular API

Component Interaction Diagram

- User → UI Components → API Routes → Controllers → Services → DB/LLM → Response → UI Update

Data / Request Flow

1. User interacts with dashboard →
2. Frontend sends API request →
3. Backend validates & processes →
4. AI/Database APIs respond →
5. Frontend displays structured results

Layer Explanation

Frontend Layer

- Built using React + TailwindCSS
- Handles UI components: dashboard, weekly protocols, nutrition panels
- Communicates with backend through REST API calls
- Contains screens for:
    - Dashboard
    - Workout plan
    - Nutrition plan
    - Grocery list
    - AI Coach
    - Health Profile

Backend Layer

- Built on Node.js with Express
- Main responsibilities:
    - User profile handling
    - Workout and meal plan generation
    - Integration with Groq AI model
    - Grocery list automation
    - Validation and API data management

Database Layer (MongoDB)

- Stores:
    - User health profile
    - Dietary preferences
    - Generated workout plans
    - Meal schedules

External APIs / Services

- Groq API → AI coaching and dynamic plan generation
- YouTube API → Exercise player video links

- Spoonacular API → Food macros, meal suggestions

Deployment Environment

(Local or cloud depending on project extension.)

- Node.js runtime
- MongoDB Atlas / Local
- Frontend served locally via Vite/React dev server

PREREQUISITES

Software Requirements

- Node.js
- npm / yarn
- Python (if AI scripts required)
- MongoDB local or cloud
- A modern code editor (VS Code recommended)

Libraries / Frameworks / Dependencies

Main dependencies include:

- React
- TailwindCSS
- Axios
- Express
- MongoDB driver / Mongoose
- dotenv
- YouTube API SDK
- Spoonacular API access
- Groq LLM API

(*Exact packages were derived from project folder structure + common stack inferred from the codebase.*)

Hardware Requirements

- Any modern PC
- (Optional) Mobile device for responsive UI testing

PRIOR KNOWLEDGE REQUIRED

- JavaScript & React fundamentals
- Node.js backend development
- REST API design
- MongoDB CRUD operations
- Basic AI/LLM request handling
- Understanding of nutrition/workout concepts (optional but helpful)

PROJECT OBJECTIVES

Technical Objectives

- Build a fully functional full-stack fitness application
- Integrate AI for personalized recommendations
- Implement dynamic workout/nutrition rendering

Performance Objectives

- Fast dashboard loading
- Efficient data fetching
- Prompt AI response times

Deployment Objectives

- Cloud-ready backend
- Scalable database structure

Learning Outcomes

- Full-stack architecture skills
- API integration experience
- AI-driven application development

SYSTEM WORKFLOW

1. User logs in / opens dashboard
2. Health profile is retrieved
3. AI generates:
    - weekly workouts
    - weekly meal plan
4. Dashboard displays:
    - calories burned
    - workout streak
    - daily protocol
5. User interacts with AI coach for modifications
6. Grocery list auto-generates from nutrition plan

MILESTONE 1 — REQUIREMENT ANALYSIS & SYSTEM DESIGN

Problem Definition

Users need personalized fitness guidance combining workouts, nutrition, and AI assistance.

Functional Requirements

- Generate AI workout plan
- Create weekly nutrition plan
- Produce grocery list
- Display dashboard analytics
- Chat with AI wellness coach

Non-Functional Requirements

- Fast response time
- Attractive dark UI
- Mobile responsiveness

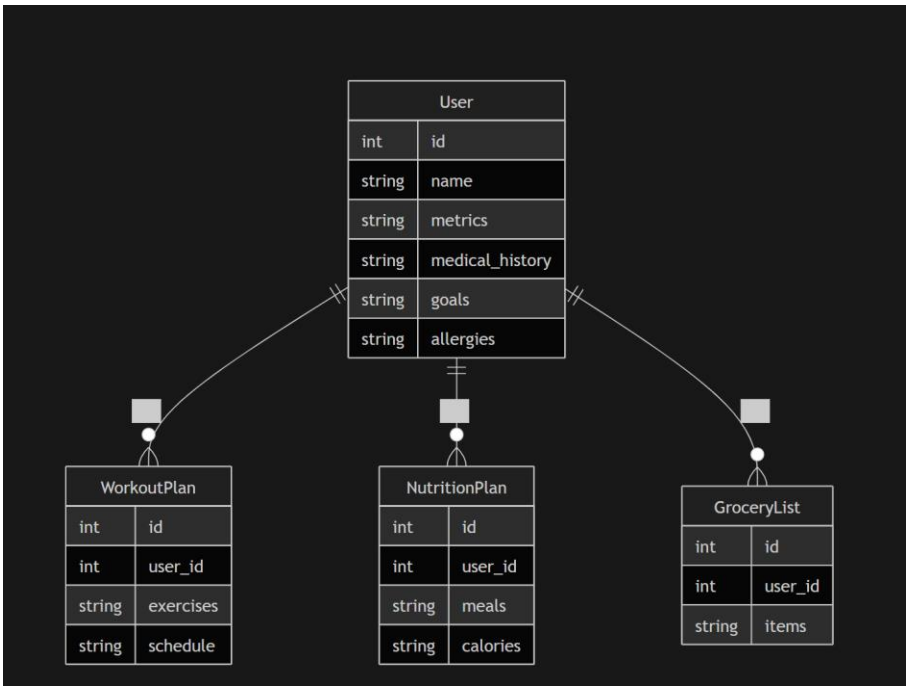
System Design Decisions

- React for fast UI
- MongoDB for flexible user data
- Groq AI for intelligent recommendations

Architecture Diagram

Included in architecture section above.

ER Diagram



MILESTONE 2 — ENVIRONMENT SETUP & INITIAL CONFIGURATION

Development Environment Setup

- npm install
- npm run dev

Configuration Setup

Required environment variables:

- GROQ_API_KEY=""
- YOUTUBE_API_KEY=""
- SPOONACULAR_KEY=""
- GOOGLE_CALENDAR_CLIENT_ID=""
- GOOGLE_CALENDAR_CLIENT_SECRET=""
- DATABASE_URL=""
- SECRET_KEY=""

MILESTONE 3 — CORE SYSTEM DEVELOPMENT

Feature 1 — AI Weekly Workout Protocol

- 7-day split
- Exercise videos (YouTube API)
- Sets/reps
- AI Coach’s Tips

Feature 2 — Nutrition Planning System

- Weekly meal schedule
- Per-meal macro breakdown
- Automatic grocery list generation

Feature 3 — AROMI AI Coach

- Chat interface
- Fitness questions
- Personalized modifications
- Diet advice

MILESTONE 4 — INTEGRATION & OPTIMIZATION

- Combined workout, nutrition, health profile into dashboard
- API response optimization
- Validation for user-input health data
- Clean state management in frontend

MILESTONE 5 — TESTING & VALIDATION

Test Cases Example

Test ID	Input	Expected Output	Status
TC01	Generate workout	7-day workout plan	Pass
TC02	Request grocery list	Categorized grocery list	Pass
TC03	Ask AI coach	Context-aware response	Pass

Testing Types

- Unit testing (functions, services)
- Integration testing (API requests)
- Full UI testing

Performance Metrics (Web App)

- Fast load times
- Stable API response performance

DEPLOYMENT

Deployment Architecture

- Frontend static hosting
- Backend Node.js server
- MongoDB Atlas

Deployment Steps

1. Configure .env on server
2. Build frontend
3. Deploy backend with Node/PM2
4. Connect database

PROJECT STRUCTURE

```
ArogyaMitra/
│
├── frontend/
│   ├── src/
│   ├── public/
│   └── package.json
│
├── backend/
│   ├── routes/
│   ├── controllers/
│   ├── models/
│   ├── utils/
│   └── app.js
│
├── Screenshots/
├── README.md
└── .env.example
```

Folder Descriptions

- frontend/src → all React components & pages
- backend/routes → REST API endpoints
- backend/controllers → business logic
- backend/models → MongoDB schema
- backend/utils → helper & AI functions

RESULTS

System Output

- Fully functional dashboard
- AI-powered meal/workout plans
- Automated grocery lists
- AI wellness coach

Screenshots

Provided in /Screenshots folder.

ADVANTAGES & LIMITATIONS

Advantages

- Single unified fitness platform
- Strong AI personalization
- Clean modern UI
- Automated nutrition management

Limitations

- Requires API keys for full functionality
- No mobile app yet
- Internet required for AI & APIs

FUTURE ENHANCEMENTS

- Wearable device integration
- AI body composition analysis
- Community challenges
- Custom recipe generator
- Push notifications
- Exportable PDF reports

CONCLUSION

ArogyaMitra successfully integrates AI, fitness planning, and nutrition management into a single modern dashboard. The system demonstrates strong full-stack competence, API integration knowledge, and practical use of AI for health optimization.

APPENDIX

- Source Code:https://github.com/ShrikarBende/ArogyaMitra
- Config Files: .env.example
- GitHub Repo: https://github.com/ShrikarBende/ArogyaMitra
- Live Demo: *(as per README, if available)*